

Instructions

Part 1: Explore the Wireless Network

In Part 1, you will explore the wireless network and verify connectivity.

Note: Wireless and cellular signal representations are **on** in this activity. However, you can turn them off by clicking **View Wireless Signals** (Ctrl+Shift+W) on the top blue toolbar.

Step 1: Explore the topology.

- a. In **Physical** mode, you will notice that **Home City** contains five different locations: a **Corporate Office**, a **Branch Office**, a **Central Office (CO)**, a **Home Office**, and a **Coffee Shop**. How are the four offices connected and what type of cables connect them?

ทุกออฟฟิศเชื่อมต่อเข้าหากันโดยวิ่งผ่านตึก **Central Office (CO)** และใช้สาย **Fiber-Optic** (ใยแก้วนำแสง) เป็นหลักในการรับส่งข้อมูลระยะไกล

- b. Navigate inside the **Corporate Office**. Notice that six Lightweight Wireless Access Points (LWAPs) are connected to the wiring closet.

How are the LWAPs connected to the network?

เชื่อมต่อตรงเข้าไปยังตู้สาขาห้องเครือข่าย (**Wiring Closet**) โดยใช้สาย **Copper Straight-Through** (สายแลนทองแดง)

- c. Navigate into the CO wiring closet.

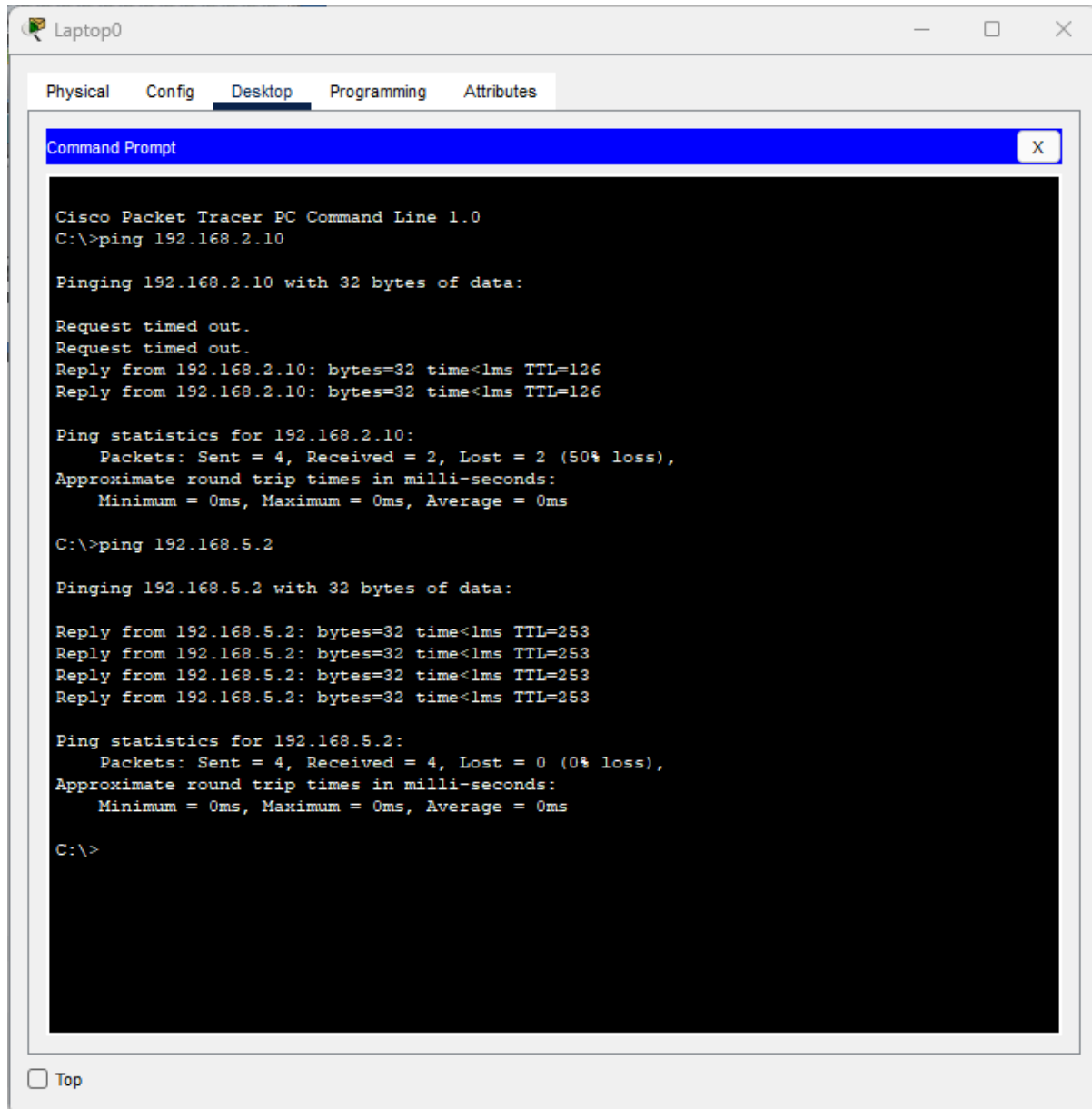
How are the cell towers connected to the **Central Office Server**?

เชื่อมต่อโดยใช้สาย **Coaxial** (สายโคแอกเชียล)

- d. Navigate to the **Branch Office**. Notice that five LWAPs are connected via copper cable to the **Branch Switch** in the wiring closet. The switch is then connected to the **Branch WLC** (wireless LAN controller).

Step 2: Verify connectivity.

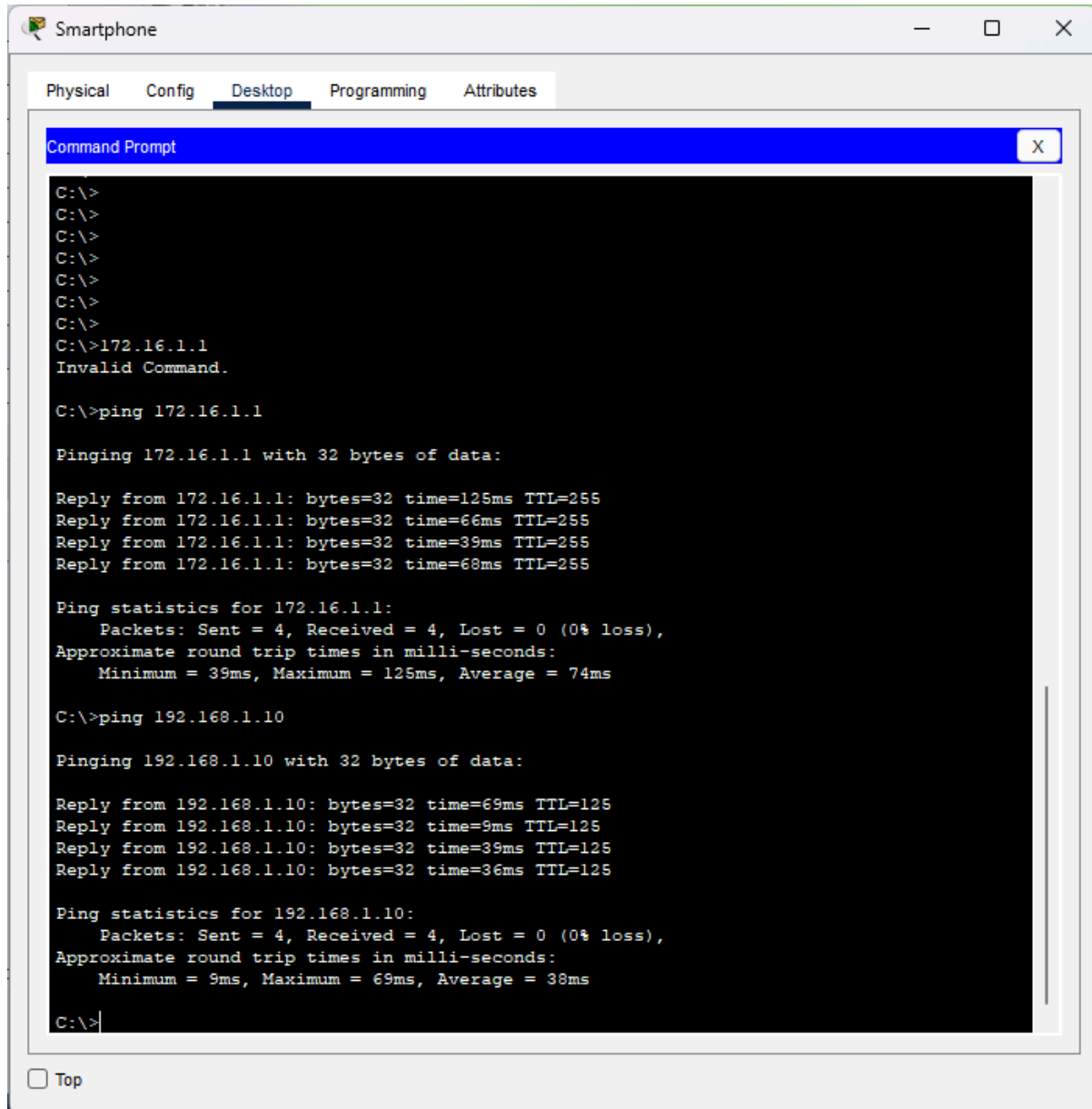
- a. To verify connectivity, navigate to **Corporate Office > Corporate Office Wiring Closet**.
- b. Click the laptop, and then **Desktop** tab > **Command Prompt**.
- 1) Ping 192.168.2.10 (the **Branch Office Server**).
 - 2) Ping 192.168.5.2 (the **Central Office** backbone connection).



Note: Packet Tracer can take some time to converge. You may receive **Request time out** messages. However, both of the pings should eventually be successful. At the bottom of the Packet Tracer interface, click **Fast Forward Time** several times to speed up convergence.

c. Navigate to the **Home City**. Click the **Smartphone** next to the cell tower above the **Central Office** building.

- d. Click **Desktop** tab, and then **Command Prompt**.
 - 1) Ping 172.16.1.1 (the cellular connection at the **Central Office**).
 - 2) Ping 192.168.1.10 (the **Corporate Office Server**).



You may receive one or two **Request time out** messages. However, both of the pings should be successful.

What are the different physical connections used between **Smartphone1** and the **Corporate Office Server**?

Smartphone1>สัญญาณไร้สาย (Cellular)>Cell Tower>สาย Coaxial>Central Office Server >สาย Fiber-Optic>Corporate Office>สาย Coppe>Corporate Server

Part 2: Add Wi-Fi Connectivity to a Boardroom

A new boardroom is being created within the **Branch Office**. Currently the boardroom is in a Wi-Fi dead zone. In Part 2, your task is to provide connectivity for devices within that room.

Step 1: Install a new LAP-PT device to provide coverage for the new boardroom.

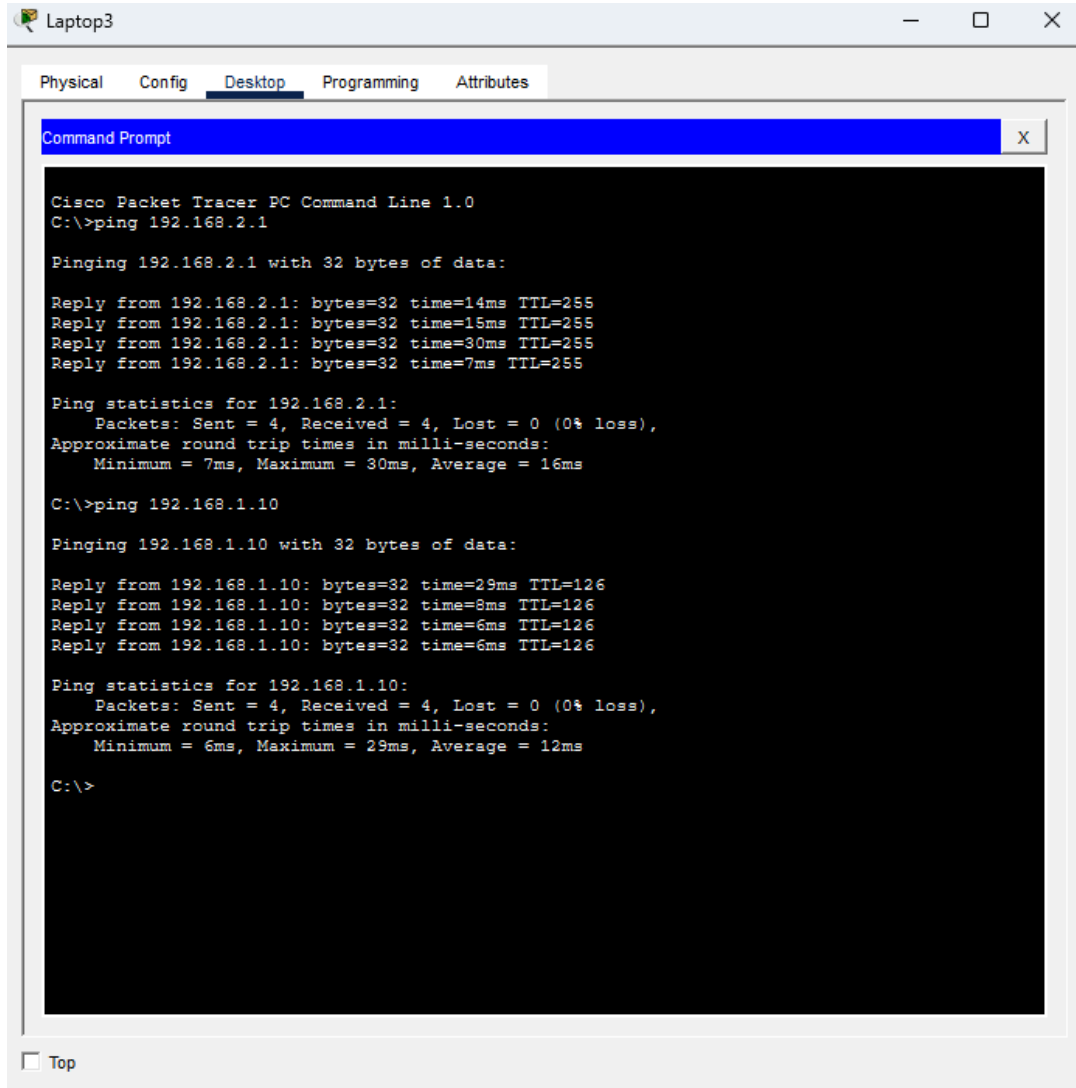
- a. Navigate to the **Branch Office**. The laptop inside the new boardroom has no access to a Wi-Fi signal.
- b. Click and drag the access point on the **Inventory Shelf** to the **Boardroom**.
- c. Click the new access point to open it. From the **Modules** menu, click and drag an **ACCESS_POINT_POWER_ADAPTER** and connect it to the power port next to the **Reset** button.
- d. Click the **Config** tab. Under **GLOBAL Settings** name the device **LAP-B6**.
- e. Under **INTERFACE Dot11Radio0** set the **Coverage Range** to **75.00** meters. Packet Tracer does not grade this setting.
- f. Close the window for **LAP-B6**. If you have **View Wireless Signals** on, you will notice that there is now coverage in the boardroom.
- g. From the **Bottom Toolbar**, click **Connections > Copper Straight-Through** cable.
- h. Connect one end of the cable to the **GigabitEthernet0** interface on the new access point. Click the rack of equipment and connect the other end of the cable to the **Rack > Branch Switch > Fa0/22 interface**.

Step 2: Verify Connectivity.

- a. In the boardroom, click the laptop, and then **Desktop** tab > **IP Configuration**. The laptop should now have full IPv4 configuration under **IP Configuration**. However, the DHCP processes may take a few minutes to complete. If necessary, toggle between DHCP and Static to resend a DHCP request. You may also need to click **Fast Forward Time** several times to speed up convergence.

b. When the IP addressing is provided, you can verify connectivity. Close the **IP Configuration** window, and then click **Command Prompt**.

- 1) Ping 192.168.2.1 (the **default gateway**).
- 2) Ping 192.168.1.10 (the **Corporate Server**).



The screenshot shows a Cisco Packet Tracer PC Command Line window titled "Laptop3". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, and a "Command Prompt" window is open. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=14ms TTL=255
Reply from 192.168.2.1: bytes=32 time=15ms TTL=255
Reply from 192.168.2.1: bytes=32 time=30ms TTL=255
Reply from 192.168.2.1: bytes=32 time=7ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 30ms, Average = 16ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=29ms TTL=126
Reply from 192.168.1.10: bytes=32 time=8ms TTL=126
Reply from 192.168.1.10: bytes=32 time=6ms TTL=126
Reply from 192.168.1.10: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 29ms, Average = 12ms

C:\>
```

At the bottom of the window, there is a "Top" button.

Part 3: Add Wireless Connectivity to a Coffee Shop in a Cellular Dead Zone

A new **Coffee Shop** has opened in the **Home City**, but there is currently no cellular service in this area. Your task is to provide cellular service for customers and employees of the coffee shop.

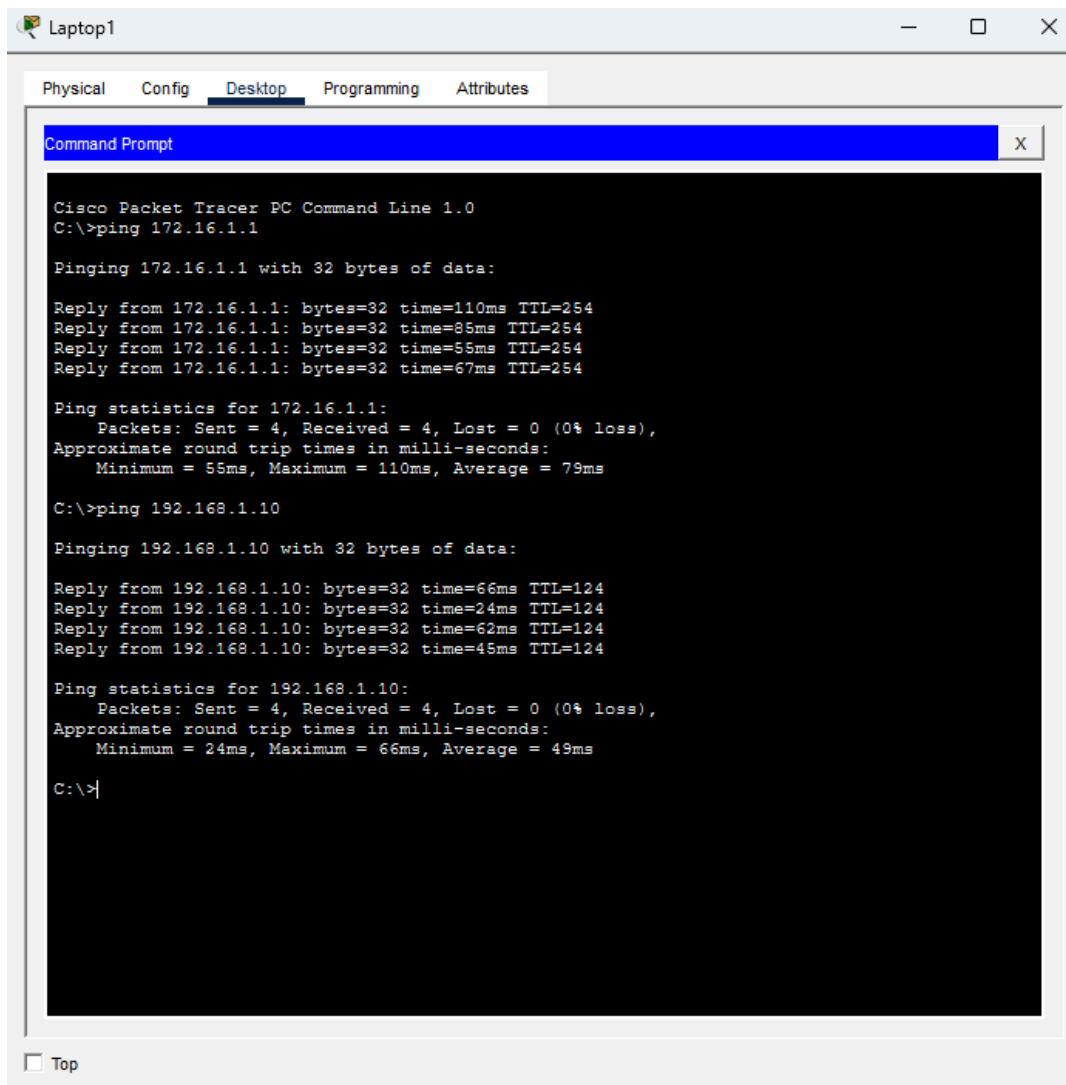
Step 1: Connect a new cell tower.

- a. Navigate to the **Home City**.
- b. Notice that the **Cell Tower** next to the **Coffee Shop** is not connected to the **Central Office**.
- c. From the Bottom Toolbar, click Connections > Coaxial cable.
- d. Connect one end to the Coaxial0 interface on the unconnected Cell Tower. Then click Central Office > Central Office Wiring Closet > Table > Central Office Server > Coaxial0/3 interface.

Step 2: Tether a laptop to the smartphone.

- a. A remote worker wants to work at the coffee shop. Navigate to the **Coffee Shop** and locate the smartphone and laptop on the table.
- b. Click **Smartphone** > **Config** tab > **3G/4G Cell1** to verify that the smartphone has received an IP address. It may take a few seconds to get addressing information. Click **DHCP Refresh**, if necessary.
- c. Click **Settings** and verify the smartphone has received a default gateway and DNS server address.
- d. Under **Cellular Tethering**, enable Bluetooth.
- e. Under **INTERFACE**, click **Bluetooth**, and set **Port Status** to **On**. Verify that **Discoverable** is enabled.
- f. In the Coffee Shop, click Laptop > Desktop tab > Bluetooth, and set Port Status to On.
- g. Select **Discover** to display **Smartphone1** under discoverable devices.
- h. Select the **Smartphone1**, click **Pair**, and then answer **Yes** to the **Bluetooth Pairing** popup message.
- i. Click the laptop again, reselect the **Smartphone1**, and then click **Tether**. It may be necessary to move the smartphone and laptop close together for Bluetooth pairing to work.
- j. After a few seconds, you should see valid addressing information in the **IP Configuration** section. If not, repeat the previous steps.

k. To verify connectivity, close the **Bluetooth Configuration** window, and then click **Command Prompt**. Ping the **cellular gateway** (172.16.1.1) and the **Corporate Office Server** (192.168.1.10). If the first ping to the **Corporate Office Server** is not successful, try another ping.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:

Reply from 172.16.1.1: bytes=32 time=110ms TTL=254
Reply from 172.16.1.1: bytes=32 time=85ms TTL=254
Reply from 172.16.1.1: bytes=32 time=55ms TTL=254
Reply from 172.16.1.1: bytes=32 time=67ms TTL=254

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 55ms, Maximum = 110ms, Average = 79ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=66ms TTL=124
Reply from 192.168.1.10: bytes=32 time=24ms TTL=124
Reply from 192.168.1.10: bytes=32 time=62ms TTL=124
Reply from 192.168.1.10: bytes=32 time=45ms TTL=124

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 66ms, Average = 49ms

C:\>
```

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Part 4: Add Wireless Connectivity to a Home Office

A remote worker for XYZ Corporation has just moved and the new house does not have a network setup yet. Your task is to set up a network to provide wireless access throughout the house and connectivity to the Corporate Office.

Step 1: Select and cable the devices.

- a. Navigate to the **Home City**, and then **Home Office**.

On the shelf behind the desk chair, there is a wireless router with external antennae. There is also a cable modem directly to the right of the wireless router. On the table in front of the couch, there is a laptop.

- b. Click **Connections > Copper Straight-Through**.
- c. Connect one end of the cable to **Port 1** of **Cable Modem**. Connect the other end to the **Internet** port on **Wireless Router**.
- d. Navigate to the **Home City** view.
- e. From the **Bottom Toolbar**, click **Connections > Coaxial** cable.
- f. Click **Home Office > Cable Modem0 > Port 0**, and then click the **Central Office > Central Office Wiring Closet > Rack > CMTS > Coaxial7** port.

Step 2: Configure the wireless router.

- a. Navigate to the **Home Office**, and click **Wireless Router > GUI** tab.
- b. The **Setup** tab is already selected. For **Internet Connection Type**, verify **Automatic Configuration - DHCP** is selected.
- c. Under **Network Setup**, verify that the following information is configured:
IP Address: **192.168.0.1**
Subnet Mask: **255.255.255.0**
DHCP: **Enabled**
Starting IP Address: **192.168.0.100**
Maximum number of users: **50**
- d. Scroll back to the top and click the **Status** tab. Under **Internet Connection**, the wireless router should have DHCP addressing from the **Central Office**. If not, click **IP Address Renew** to resend DHCP messages.
- e. Click the **Wireless** tab.
- f. Configure the 2.4 GHz network with **homesweethome** as the **Network Name (SSID)**. Scroll to the bottom and click **Save Settings**.
- g. Scroll back to the top and select the **Wireless Security** subtab.
- h. For the 2.4 GHz **Security Mode**, select **WPA2-Personal**, and then configure **mySecureNet** as the **Passphrase**. Scroll to the bottom and click **Save Settings**.

Step 3: Verify connectivity.

- On the table in front of the couch, click the laptop, and then the **Config** tab. Next, select **Wireless0** under **INTERFACE**.
- Enter the SSID **homesweethome**.
- Select **WPA2-PSK** for the **Authentication** method, and then configure **mySecureNet** as the **PSK Pass Phrase**.
- Under **IP Configuration**, the laptop should receive DHCP addressing. You may need to toggle between DHCP and Static a few times to resend DHCP requests.
- Click **Desktop** tab > **Command Prompt**. Ping various addresses throughout the network. For example, the following pings should be successful:

The screenshot shows the 'Laptop2' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'GLOBAL' and 'INTERFACE' expanded. Under 'INTERFACE', 'Wireless0' is selected. The main panel displays the configuration for 'Wireless0'.

Wireless0

Port Status: ☒ On

Bandwidth: 200 Mbps

MAC Address: 0090.2BE8.CA46

SSID: homesweethome

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK ☐ WPA ☐ 802.1X

WEP Key:

PSK Pass Phrase: mysecureNet

User ID:

Password:

Method: MD5

User Name:

Password:

Encryption Type: AES

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.0.100

Subnet Mask: 255.255.255.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address:

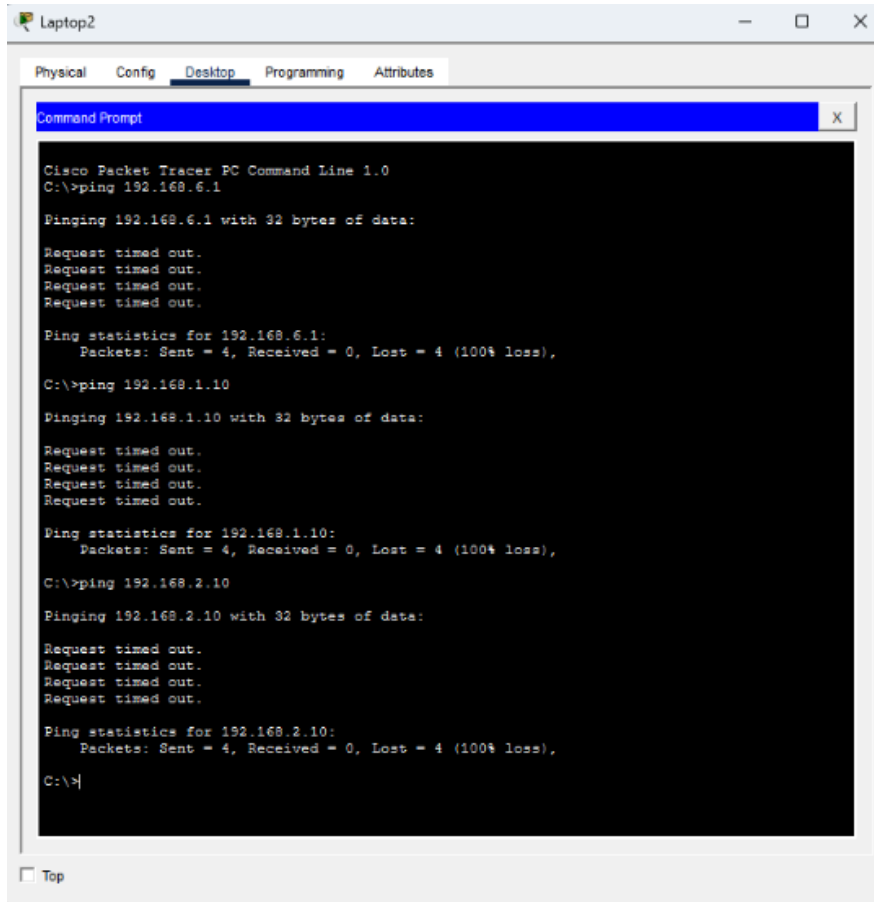
Link Local Address: FE80::290:2BFF:FEE8:CA46

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ping 192.168.6.1 (**Central Office router G1/0**)

ping 192.168.1.10 (**Corporate Office Server**)

ping 192.168.2.10 (**Branch Office Server**)



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.6.1

Pinging 192.168.6.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.6.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

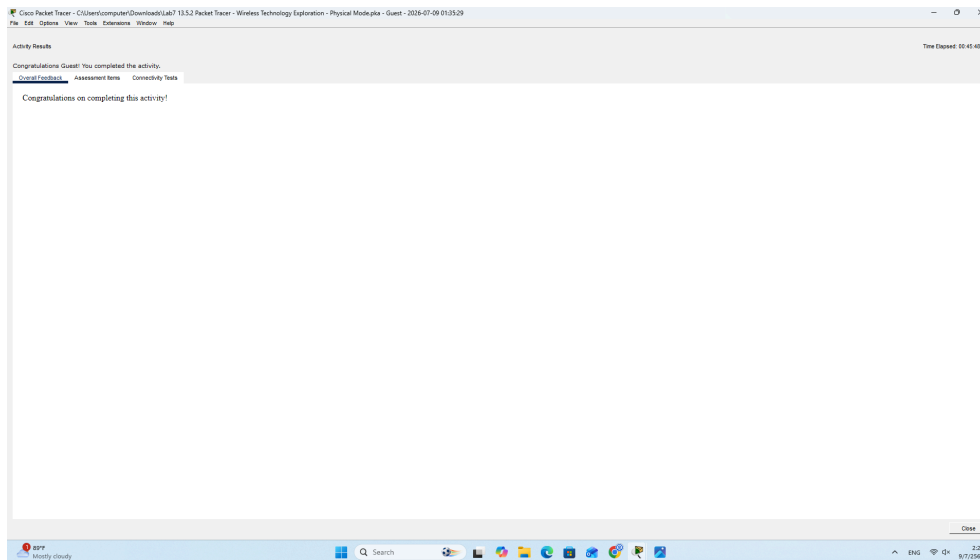
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



Reflection

1. What overall benefit does wireless technology provide to the end user?

ให้ความสะดวกสบายและความยืดหยุ่นสูง (Mobility) ผู้ใช้งานสามารถเคลื่อนย้ายอุปกรณ์ไปทำงานตามจุดต่างๆ ได้โดยไม่ต้องพะวงเรื่องการเดินสายสัญญาณ และสามารถเข้าถึงข้อมูลสารสนเทศได้ทุกที่ทุกเวลา

2. Which form of wireless networking is the most beneficial?

ไม่มีรูปแบบไหนดีที่สุดเพราะขึ้นอยู่กับการใช้งาน หากเป็นการใช้งานภายในอาคารหรือพื้นที่ปิด **Wi-Fi** จะได้เปรียบเรื่องความเร็วที่สูงและไม่มีค่าบริการข้อมูลเพิ่ม แต่หากเป็นการเดินทางหรืออยู่นอกอาคารในพื้นที่กว้าง เครือข่ายโทรศัพท์มือถือ (**Cellular เช่น 4G/5G**) จะมีประโยชน์สูงสุดเนื่องจากครอบคลุมพื้นที่ได้กว้างไกลกว่า

3. How could each of the following groups benefit from the various wireless technologies presented in this activity?

- Student

สามารถใช้ Wi-Fi ของสถาบันเพื่อสืบค้นข้อมูล ส่งงาน หรือนั่งเรียนออนไลน์จากที่พักใดก็ได้ในมหาวิทยาลัย หรือใช้ Cellular ทำงานกลุ่มนอกสถานที่ได้สะดวก

- Salesperson

สามารถใช้เทคโนโลยี Cellular หรือการเชื่อมต่อสปีดเพื่อเช็คสต็อกสินค้า นำเสนอแผนงาน หรือส่งใบสั่งซื้อ กลับเข้าเซิร์ฟเวอร์บริษัทได้ทันทีในขณะที่ออกไปพบลูกค้านอกสถานที่

- R&D Engineer

สามารถเชื่อมต่ออุปกรณ์หรือเครื่องมือทดสอบแบบไร้สายเพื่อเก็บข้อมูลแลกเปลี่ยนการทดลองในห้องแล็บได้อย่างคล่องตัว ไม่เกาะสายไฟ และทำงานร่วมกับทีมผ่านระบบ Cloud ได้ง่ายขึ้น

- Corporate Executive

สามารถเข้าถึงระบบบริหารจัดการส่วนกลาง (Dashboard) หรือเข้าร่วมการประชุมทางไกล (Video Conference) ได้จากทุกมุมโลก ไม่ว่าจะอยู่ที่บ้าน ร้านกาแฟ หรือระหว่างการเดินทาง เพื่อให้สามารถตัดสินใจในเรื่องสำคัญขององค์กรได้ทันเวลาที่